

Program	: Computer Engineering Program	Date	: 8 March 2026
Examiners	: Dr. Tarek salem Eng :Basant , Eman, Habiba	Duration	2 hours
Course/Code	: Database Management Systems / CSE112		

Part 1: Quick Summary

Q: What is a Data Model?

A set of Concepts used to describe **database structure + rules (constraints)**.

Includes:

- Structure → tables, attributes
- Relationships → links between data
- Constraints → rules for valid data

Q: What are the categories of data models?

Conceptual Data Model

High-level → describes **meaning of data** (entities, relationships)

Implementation Data Model

Middle level → how DBMS represents data

Physical Data Model

Low-level → describes **how data is stored** in memory/files

Schema vs Instance

Q: What is Database Schema?

A: The structure of the database including tables, attributes and constraints.

Q: What is Database Instance (State)?

A: The actual data stored in the database at a specific time.

Database Schema -: Database **structure/design**

(tables, attributes, constraints)

Rarely changes.

Database Instance (State) -:Actual **data stored at a specific time**Changes frequently.

Schema = **Intension**

Instance = **Extension**

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● Example of a Database Schema:

STUDENT

Name	Student_number	Class	Major
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COURSE

Course_name	Course_number	Credit_hours	Department
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PREREQUISITE

Course_number	Prerequisite_number
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SECTION

Section_identifier	Course_number	Semester	Year	Instructor
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GRADE_REPORT

Student_number	Section_identifier	Grade
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Figure 2.1

Schema diagram for the database in Figure 1.2.

Three Schema Architecture

Q: What are the three levels of database architecture?

External Level

User View

Different users see different data.

Conceptual Level

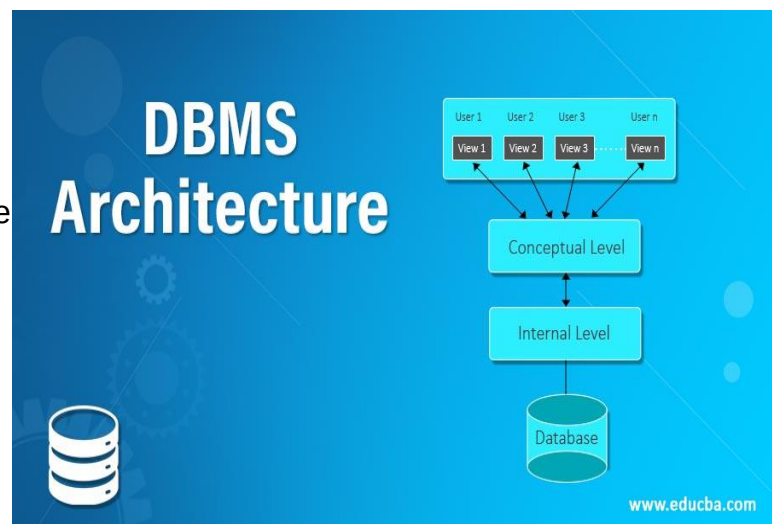
Logical structure of the whole database

Entities – attributes – relationships.

Internal Level

Physical storage of data

Files – indexes – storage details.



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DBMS Languages

Q: What is DDL?

A: Language used to define the database structure.

Examples:

CREATE, ALTER, DROP

Q: What is DML?

A: Language used to manipulate data.

Examples:

SELECT, INSERT, UPDATE, DELETE

Types of DML

Q: What are the types of DML?

High-Level DML (Non Procedural)

User specifies **WHAT data is needed**

Example: SQL

Low-Level DML (Procedural)

User specifies **HOW to access data step by step**

Data Dictionary

Q: What is Data Dictionary?

A: A repository that stores metadata about the database.

Definition:

Central repository storing **metadata about database**.

Metadata = **data about data**

Contains:

Table names
Attribute names

Data types
Constraints

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Types of Data Dictionary

Active Data Dictionary

Integrated with DBMS
Automatically updated.

Passive Data Dictionary

Not integrated with DBMS
Updated manually.

DBMS Architecture

Q: What are DBMS architectures?

Centralized DBMS

All processing done in **one central system**.

Client-Server Architecture

Clients request services
Server processes database queries.

Two-Tier Client–Server Architecture

Clients send requests → **Servers process** data/services.

Specialized Servers

Servers perform **specific tasks**.

Examples:

- **Print Server** → printing
- **File Server** → file storage
- **DBMS Server** → database management
- **Web Server** → web services
- **Email Server** → email handling

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Clients access these servers **when needed**.

Clients

Computers that **request services from servers**.

Characteristics:

- Provide **user interface**
- Can be **PCs / workstations / diskless machines**
- Connected through **network (LAN / wireless)**

Three Tier Architecture

Client → Application Server → Database Server

Improves:

- Security
- Scalability
- Performance

What are the Key Differences Between Centralized and Client-Server DBMS

Feature	Centralized DBMS	Client-Server DBMS
Data Storage	Single server stores everything	Database stored on a central server, accessed by clients remotely over a network.
Processing	Processing and storage happen on one machine	Processing is split between client and server
Access	Direct access from terminals	Clients send queries to the server
Scalability	Limited (performance decreases as users increase)	Highly scalable (supports many users)

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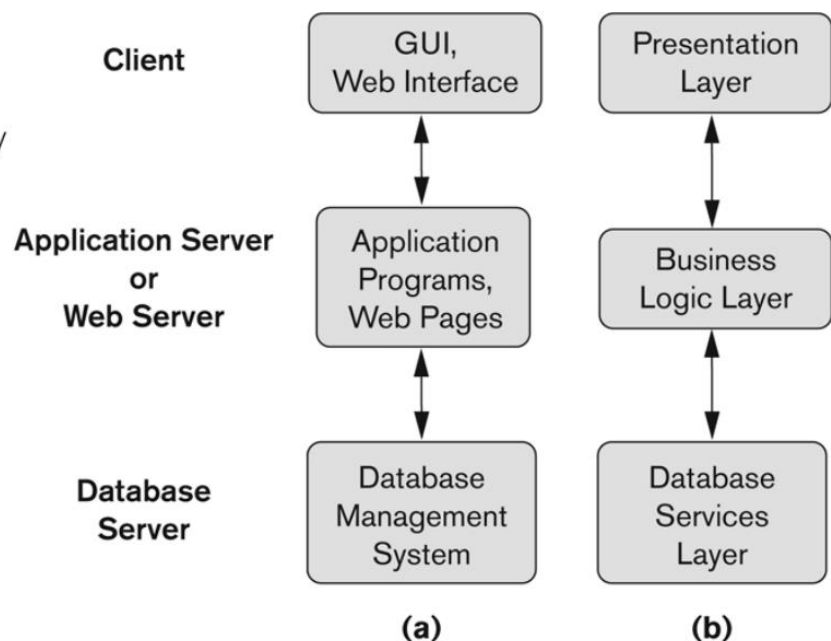
Compare and contrast 2-tier and 3-tier client-server architectures.

A **2-tier architecture** consists of **clients and specialized servers**, where clients directly access **print, file, DBMS, web, or email servers**. It is **simpler** but lacks **scalability and security** due to direct database access.

A **3-tier architecture** introduces an **application server (middle tier)** between clients and the database, handling **business logic and security**. It improves **scalability, security, and performance** by preventing direct client access to the database.

Figure 2.7

Logical three-tier client/server architecture, with a couple of commonly used nomenclatures.



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Part 2:- Questions

I. Multiple Choice Questions (MCQ) – 20

- A data model is used to:**
 - Store programs
 - Describe database structure and rules
 - Control hardware
 - Manage networks
- Which data model focuses on the meaning of data?**
 - Physical
 - Conceptual
 - Implementation
 - Internal
- Which data model describes how data is stored in memory?**
 - Conceptual
 - Logical
 - Physical
 - External
- Database schema represents:**
 - Current data in database
 - Database structure
 - Query results
 - Application programs
- Database instance refers to:**
 - Database structure
 - Database schema
 - Actual data stored at a specific time
 - Data model

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6. Schema is also called:

- a) Extension
- b) Instance
- c) **Intension**
- d) Snapshot

7. The database state is also called:

- a) Intension
- b) **Extension**
- c) Schema
- d) Structure

8. Which level represents user view of the database?

- a) Internal
- b) Conceptual
- c) **External**
- d) Physical

9. Which level describes logical structure of the database?

- a) External
- b) **Conceptual**
- c) Internal
- d) Physical

10. Which level describes physical storage of data?

- a) External
- b) Conceptual
- c) **Internal**
- d) Logical

11. DDL is used to:

- a) Retrieve data
- b) **Define database structure**
- c) Update records
- d) Delete data

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12. Which command belongs to DDL?

- a) SELECT
- b) INSERT
- c) **CREATE**
- d) UPDATE

13. Which command belongs to DML?

- a) DROP
- b) CREATE
- c) **SELECT**
- d) ALTER

14. High-level DML means:

- a) Specify HOW to get data
- b) **Specify WHAT data is needed**
- c) Used only by DBA
- d) Used for storage

15. Low-level DML means:

- a) **Specify HOW to access data**
- b) Specify WHAT data is needed
- c) Used for backup
- d) Used for schema design

16. Data dictionary stores:

- a) Actual data
- b) **Metadata**
- c) Programs
- d) Queries

17. Active data dictionary is:

- a) Updated manually
- b) **Integrated with DBMS**
- c) Used for backup
- d) Used for queries

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18. Which architecture processes everything in one system?

- a) Client-server
- b) **Centralized**
- c) Distributed
- d) Three-tier

19. In two-tier architecture:

- a) Clients communicate through application server
- b) **Clients directly access server**
- c) Database is local
- d) No server exists

20. Three-tier architecture includes:

- a) Client and server only
- b) Client – Database – Network
- c) **Client – Application Server – Database Server**
- d) Client – File Server – Printer

True / False

- 1. A data model describes database structure and rules. **True**
- 2. Database schema changes frequently. **False**
- 3. Database instance represents actual data at a specific time. **True**
- 4. External level describes physical storage. **False**
- 5. Conceptual level describes logical database structure. **True**
- 6. DDL is used to manipulate data. **False**
- 7. SELECT command belongs to DML. **True**
- 8. Data dictionary stores metadata about the database. **True**
- 9. Client-server architecture separates clients from servers. **True**
- 10. Three-tier architecture improves security and scalability. **True**

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Complete the Following

1. A _____ is a set of concepts used to describe database structure.
Data Model
2. The structure of the database is called _____.
Schema
3. The actual data stored in the database is called _____.
Instance
4. The user view of the database is represented by the _____ level.
External
5. The logical structure of the database is represented by the _____ level.
Conceptual
6. The physical storage of data is represented by the _____ level.
Internal
7. _____ is used to define database structure.
DDL
8. _____ is used to retrieve and manipulate data.
DML
9. A repository storing metadata is called _____.
Data Dictionary
10. The architecture that includes an application server is _____.
Three-tier architecture

Short Answer Questions

1. Define Data Model.
2. What are the categories of data models?
3. Define Database Schema.
4. Explain the three schema architecture.
5. What is DDL?